Gradient Boosting:

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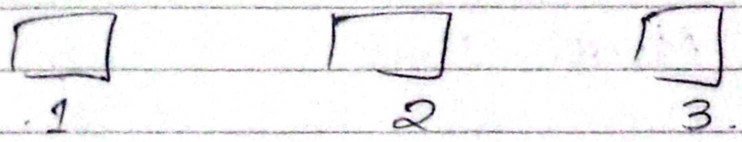
Intuition.



1st model - regressor [mean of all output]
independent of input value.

error

ig	cgpa	salary	pred 1	res1	pred2	res2	pred3
90	8	3	4.8	-1.8	-1.8	-1.62	-1.62
100	7	4	4.8	-0.8	-0.8	-0.72	-0.72
110	6	8	4.8	3.2	3.2	2.88	2.88
120	9	6	4.8	1.2	1.2	1.08	1.08
80	5	3	4.8	-1.8	-1.8	-1.62	-1.62

For another model, it gives loss f^n's error.

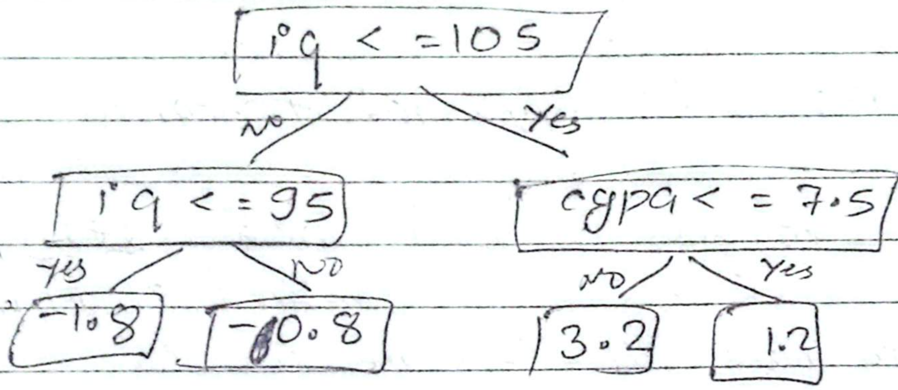
loss $f^n \Rightarrow$ actual - predicted = pseudo residual

Now, we have to send error of model 1 to model 2.

Generally, our model 2 will be our decision tree

For model 2,
input colⁿ: iq cgpa   Output colⁿ: res1.

- Model 2 should ~~calculate~~ predict the error done by model 1.



# If there is 2 models only, then:
pred = output of model 1 + model 2.
(m1) + (m2)

⇒ It will be overfitting.
So; Output = $\hat{y}(m_1 + \eta m_2)$

learning rate ~ 0.1.

$= 4.8 - 0.1 \times 1.8$

$= 4.7.$

- If this process continue with other models, then we will get better answer.

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res 2 :

actual - pred

$3 - (\underbrace{4.8}_{\text{pred1}} + \underbrace{0.1}_{lr} \times (\underbrace{-1.8}_{\text{pred2}}))$

$= -1.62$

res will approach towards zero.

- Now, model 3 will pred $\underbrace{\text{res 2}}_{\text{combined error of m1 \& m2}}$
- Learning rate will be same for all.

res 3 :

~~3 = (-1.8~~

$y\text{-}pred = m1 + 0.1 \times m_2 + 0.1 \times m_3$

$= \cancel{4.8}\ - 0.18 - 0.162$

4.5

Adaboost	Gradient descent.
• Decision stump max_depth = 1.	• Decision tree max_depth = 8 - 32
• weight for diff model	• learning rate same for all model.